

3.2. Singular Deformations

According to 2.7, the versal moduli space of deformations of a singularity of a supercurve is a subspace of the versal moduli space of the underlying superscheme. To first order, one has the following result.

Lemma 3.3. The versal moduli space of a singularity of a supercurve is, to first order, the following subspace of the versal moduli space, computed in 3.1, of the superscheme singularity:

$$\begin{aligned} \text{I: } & \tau = 0, \\ \text{II: } & \tau_1 = \tau_2 = 0, \quad t_1 + t_2 = 0. \end{aligned}$$

The point is to know when one can deform δ , while respecting the following conditions:

- (a) δ takes values in ω_X ;
- (b) on the smooth locus, δ satisfies 1.4(ii).

In the deformations under consideration, parametrized by a vector subspace of the one that parametrizes the deformation of 3.1, the complement U of the singular point is the union of two branches. On these branches we take the respective coordinates (z_1, θ_1) and (z_2, θ_2) . We describe a section ω of ω_X by its restrictions ω_1 and ω_2 to the two branches. The condition on ω_1 and ω_2 is that, for every section f of $\mathcal{O}_X$, with restrictions f_1 and f_2 to the two branches, one has in $\mathcal{O}_S$

$$\text{Res}_0(f_1\omega_1) + \text{Res}_0(f_2\omega_2) = 0.$$

Since ω_X is flat over S , to obtain a set of generators of ω_X it is enough to lift a set of generators of ω_{X_0} , where X_0 is the special fibre.

In case I, respectively II, let b denote the following section of ω_X on U :

$$\begin{aligned} \text{I: } & \begin{cases} (1) \left[\frac{dz_1/z_1}{d\theta} \right], \\ (2) \left[-\frac{dz_2/z_2}{d\theta} \right], \end{cases} \\ \text{II: } & \begin{cases} (1) \left[\frac{dz_1}{d\theta_1} \right], \\ (2) \left[\frac{dz_2}{d\theta_2} \right]. \end{cases} \end{aligned}$$

Let D_0 be the odd derivation on U :

$$\begin{aligned} \text{I: } & \begin{cases} (1) \partial_\theta + \theta \partial_{z_1}, \\ (2) \partial_\theta - \theta \partial_{z_2}, \end{cases} \\ \text{II: } & \begin{cases} (1) \partial_{\theta_1} + \theta_1 \partial_{z_1}, \\ (2) \partial_{\theta_2} + \theta_2 \partial_{z_2}. \end{cases} \end{aligned}$$

Set $\delta_0 = D_0 \cdot b$. This δ_0 does not necessarily send $\mathcal{O}_X$ to ω_X ; for a deformation as a supercurve to be possible, one must be able to correct δ_0 so that it sends $\mathcal{O}_X$ to ω_X .

We distinguish cases I and II, and the even and odd deformations.

Case I, odd. The deformation is

$$z_1 z_2 = \tau \theta.$$

Here b lies in ω_X ; one has

$$b = \frac{[dz_1 dz_2 / d\theta]}{d(z_1 z_2 - \tau \theta)},$$

so it is a basis of ω_X . One would have to correct the odd derivation D_0 , that is, pass to $D_0 + \tau D$, where D is an even derivation with values in $\mathcal{O}$. Describing a section of $\mathcal{O}_X$ by its restrictions to the branches, one has

$$z_1 = \begin{cases} (1) & z_1, \\ (2) & \tau \theta / z_2 \end{cases} \xrightarrow{D_0} \begin{cases} \theta z_1, \\ \tau / z_2, \end{cases}$$

which is not in $\mathcal{O}_X$: θz_1 is in $\mathcal{O}_X$, while τ / z_2 is not, since 0 is not in $(1/z_2)$. The derivation D would have to send z_1 to a nonzero quantity on the second branch; this is impossible.

Case II. Absence of odd deformations. The X_0 is bihomogeneous; one can assign the weights

$$\theta_1 : (1, 0), \quad z_1 : (2, 0), \quad \theta_2 : (0, 1), \quad z_2 : (0, 2).$$

The versal deformation of the singularity is also bihomogeneous:

$$\tau_1 : (1, 2), \quad \tau_2 : (2, 1), \quad t_1, t_2 : (1, 1).$$

Therefore, if there is a deformation with

$$\tau_1 = a\tau, \quad \tau_2 = b\tau, \quad (a, b) \neq (0, 0),$$

there is also one with (a, b) replaced by $(\lambda a, \mu b)$. If a or b is zero, then by symmetry there is a deformation with $\tau_2 = 0$; otherwise, there are two distinct deformations and τ_1, τ_2 are in fact free.

It is therefore enough to study the deformation over $k[\tau_1]$:

$$z_1 z_2 = \tau_1 \theta_1, \quad z_1 \theta_2 = z_2 \theta_1 = \theta_1 \theta_2 = 0.$$

We compute ω_X . Write an element of ω_X according to its values on each branch:

$$\begin{pmatrix} u_1 \left[\frac{dz_1}{d\theta_1} \right] \\ u_2 \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix}.$$

One has to lift the generators. The generator $[dz_1/d\theta_1]$ is not, as it stands, in ω_X , because

$$z_2 \left[\frac{dz_1}{d\theta_1} \right] = \begin{cases} \frac{\tau_1 \theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right], \\ z_2 \cdot 0, \end{cases}$$

has residue τ_1 . It lifts to

$$\begin{cases} \left[\frac{dz_1}{d\theta_1} \right], \\ -\frac{\tau_1 \theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right]. \end{cases}$$

Similarly, $[dz_2/d\theta_2]$ lifts to

$$\begin{cases} 0, \\ \left[\frac{dz_2}{d\theta_2} \right]. \end{cases}$$

Finally,

$$\frac{\theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right] - \frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right]$$

lifts by the same expression on both branches.

Now seek to lift

$$\delta = D_0 \cdot b$$

as in 3.1, II. The rational lift given by the same formula on each branch is not valued in ω_X : one has

$$\theta_1 = \begin{cases} \theta_1, \\ 0 \end{cases} \mapsto \left[\frac{dz_1}{d\theta_1} \right].$$

The derivation D_0 would have to be corrected by an even rational derivation sending

$$\theta_1 \mapsto \frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right]$$

plus a regular section on the second branch, which is impossible.

Case II, even deformations. One has

$$\begin{aligned} z_1 z_2 &= 0, & \theta_1 \theta_2 &= 0, \\ z_1 \theta_2 &= t_1 \theta_1, & z_2 \theta_1 &= t_2 \theta_2. \end{aligned}$$

On the two branches:

branch 1: $z_2 = 0, \quad \theta_2 = t_1 \theta_1 / z_1, \quad \text{coordinates } (z_1, \theta_1),$

branch 2: $z_1 = 0, \quad \theta_1 = t_2 \theta_2 / z_2, \quad \text{coordinates } (z_2, \theta_2).$

We again look for generators of ω_X , in the form

$$\begin{cases} u_1(z_1, \theta_1) \left[\frac{dz_1}{d\theta_1} \right], \\ u_2(z_2, \theta_2) \left[\frac{dz_2}{d\theta_2} \right], \end{cases} \quad u_i \in k[z_i, z_i^{-1}; \theta_i].$$

The generator $[dz_1/d\theta_1]$ lifts as

$$\begin{pmatrix} \left[\frac{dz_1}{d\theta_1} \right] \\ -\frac{t_1}{z_2} \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix},$$

and $[dz_2/d\theta_2]$ lifts as

$$\begin{pmatrix} -\frac{t_2}{z_1} \left[\frac{dz_1}{d\theta_1} \right] \\ \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix}.$$

Moreover,

$$\begin{pmatrix} \frac{\theta_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right] \\ -\frac{\theta_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix}$$

lifts in the same way.

Let D_0 be as above. One has

$$D_0(z_1) = \begin{pmatrix} \theta_1 \left[\frac{dz_1}{d\theta_1} \right] \\ 0 \end{pmatrix}, \quad D_0(z_2) = \begin{pmatrix} 0 \\ \theta_2 \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix},$$

which are in ω_X . Next

$$D_0(\theta_1) = \begin{pmatrix} \left[\frac{dz_1}{d\theta_1} \right] \\ \frac{t_2}{z_2} \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix}, \quad \text{is in } \omega_X \text{ if } t_1 = -t_2,$$

and

$$D_0(\theta_2) = \begin{pmatrix} \frac{t_1}{z_1} \left[\frac{dz_1}{d\theta_1} \right] \\ \left[\frac{dz_2}{d\theta_2} \right] \end{pmatrix}, \quad \text{is in } \omega_X \text{ if } t_1 = -t_2.$$

If $t_1 + t_2 \neq 0$, one must correct D_0 to $D_0 + \varepsilon D$, with D even; once again, D would have to send θ_1 to a nonzero quantity on the second branch, which is impossible.

Thus one can deform only when

$$t_1 + t_2 = 0.$$

That the non-excluded deformations are possible follows from 2.5.